



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PURE MATHEMATICS
PAPER 2

6042/2
3 hours

NOVEMBER 2025 SESSION

Additional materials:
Answer booklet
List of Formulae MF 7
Scientific calculator [Non-programmable]

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Answer **all** questions in Section A and any **five** questions from Section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 6 printed pages.

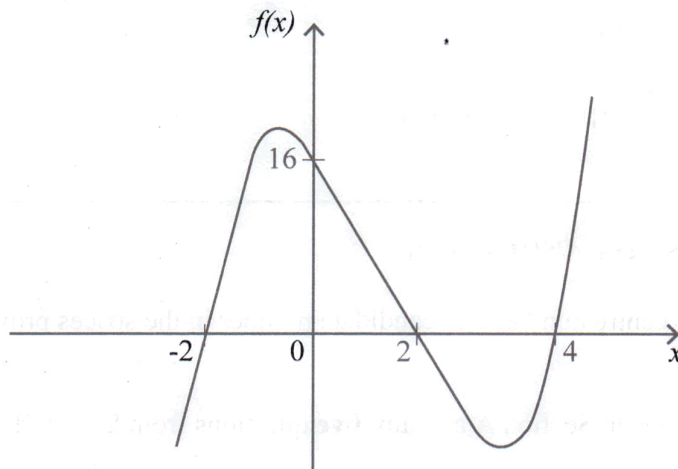
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Section A (40 marks).

Answer all questions in this section.

- 1 (a) Sketch on the same axes, the graphs of $y = |x^2 - 2|$ and $y = -x$. [3]
- (b) Hence or otherwise find the range of values of x for which $|x^2 - 2| < -x$. [3]
- 2 The diagram shows the graph of $y = f(x)$.



Sketch on separate diagrams, showing clearly all the intercepts, if any, the graphs of

- (a) $y = f(x) - 6$, [1]
- (b) $y = f(x + 1)$, [2]
- (c) $y = -\frac{1}{2}f(x)$, [2]
- (d) $y = f(2x)$. [2]
- 3 (a) By differentiating from first principles, find the gradient function of the function

$$f(x) = \frac{2x-3}{x^2}. \quad [5]$$

- (b) Hence, or otherwise, find the equation of the tangent to the curve $f(x) = \frac{2x-3}{x^2}$ at the point where $x = 2$. [3]

- 4 The complex number u is given by $u = \frac{5+4i}{3-2i}$

- (a) Express u in the form $x + iy$, where x and y are real numbers. [2]
- (b) Hence or otherwise, find the modulus and argument of u . [4]



(c) Sketch an Argand diagram showing the point representing the complex number u . [2]

(d) Show on the same axes, the locus of the complex number z such that $|z - u| = 2$. [2]

5 The function $f: x \rightarrow \frac{m}{x} + n$, where m and n are constants, is such that $f(-1) = 1\frac{1}{2}$ and $f(2) = 9$.

(a) State the domain of $f(x)$. [1]

(b) Find the value of m and n . [4]

(c) Evaluate $f^{-1}(4)$. [4]



Section B (80 marks)

Answer any **five** questions from this section. Each question carries **16** marks.

- 6 (a) Solve the inequality

$$\frac{2}{x-3} < \frac{3}{x-4}. \quad [5]$$

- (b) If $(1; 5)$ is a solution of the equations

$$\begin{aligned} n^2x + my &= 14 \\ ny - mx &= 8 \end{aligned}$$

find the values of n and m . [5]

- (c) (i) Sketch on a single diagram the graphs of

$$y = 5 - |x| \text{ and } y = |2x - 3|. \quad [3]$$

- (ii) Hence, or otherwise, solve the inequality $|2x - 3| < 5 - |x|$. [3]

- 7 (a) (i) Find the equation of the curve which satisfies the differential equation

$$\frac{dy}{dx} = \frac{-x}{y}$$

and passes through the point $(-4; 3)$. [4]

- (ii) Hence describe the solution curve. [1]

- (b) The rate of destruction of drug by the kidney is proportional to the amount of the drug present in the body. At time, t , hours, the quantity of the drug in the body is x mg.

- (i) Form a differential equation relating x and t . [2]

- (ii) Show that the general solution of the differential equation is

$$x = Ae^{-kt} \text{ where } A \text{ is a constant.} \quad [3]$$

- (iii) At $t = 0$, a quantity of 200 mg of the drug is administered.

When $t = 1$ hour, the amount of the drug in the body is 180 mg.

Find the quantity of the drug in the body after 8 hours. [6]



- 8 (a) The n^{th} term of a series is given by

$$4^{n-4} + 6n.$$

Find the sum of the first N terms.

- (b) (i) Find Maclaurin's series for

$$f(x) = (1 - x)^{-1}$$

up to and including the term in x^4 .

- (ii) Find the expansion of

$$(9 - x)^{\frac{3}{2}}(1 + 3x)^{\frac{3}{2}}$$

up to and including the term in x^2 .

- (iii) Hence find the set of values of x for which the expansion in (ii) is valid.

- 9 (a) Express $2\cos\theta - \sqrt{3}\sin\theta$, in the form $R\sin(\theta - \alpha)$.

- (b) Solve the equation $2\cos\theta - \sqrt{3}\sin\theta = 1$.

- (c) Hence sketch the graph of $y = 2\cos\theta - \sqrt{3}\sin\theta$, for $0^\circ \leq \theta \leq 360^\circ$

- (d) State the maximum and minimum values of $2\cos\theta - \sqrt{3}\sin\theta$, for $0^\circ \leq \theta \leq 360^\circ$.

- (e) Express

$$2\cos^2\theta - 1$$

in terms of $\cos 2\theta$.

- 10 (a) Find an appropriate value of

$$\int_0^1 x^2 e^x dx$$

using the trapezium rule with 5 ordinates.

- (b) Evaluate the exact value of

$$\int_0^1 x^2 e^x dx$$

using integration by parts.



- (c) Hence state whether the approximation obtained in (a) is an over-estimation or under-estimation. [1]

- (d) Use Newton-Raphson method to obtain a root of the equation

$$x^3 + 3x - 7 = 0$$

given that $x_0 = 1$. Give answer correct to 2 d.p. [5]

- 11 (a) An error of 3% is made in measuring the radius of a sphere.

Find the percentage error in the volume. [volume of sphere = $\frac{4}{3}\pi r^3$.] [6]

- (b) (i) Express

$$\frac{2x+1}{(x-2)(x^2+1)}$$

in partial fractions. [6]

- (ii) Hence, show that

$$\int_0^1 \frac{2x+1}{(x-2)(x^2+1)} dx = -\frac{1}{2} \ln 8. [4]$$

- 12 (a) Consider the table below for the set $R = (\{0; N; I; T\}r)$.

r	O	N	I	T
O	O	N	I	T
N	N	I	T	O
I	I	T	O	N
T	T	O	N	I

Show that $R = (\{0; N; I; T\}r)$ forms a group. [6]

- (b) (i) Determine the single matrix M that describes an enlargement by factor 2 followed by a shear, factor 3 in the x-axis direction with y-axis invariant. [4]

- (ii) Triangle ABC, with coordinates A(3;2), B(1;1) and C(5;1) is transformed using matrix M in (b)(i).

Find the coordinates of A^1 , B^1 and C^1 , the images of points A, B and C. [4]

- (iii) Find the area of triangle $A^1B^1C^1$. [2]

